

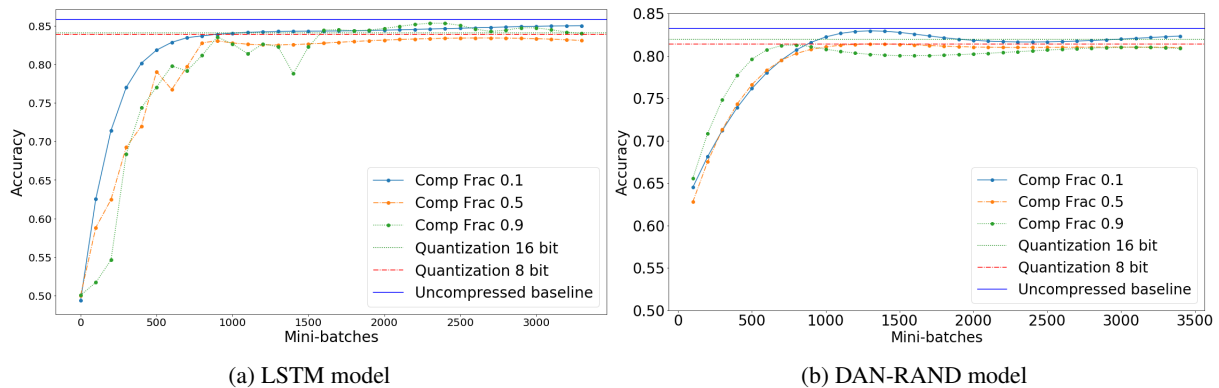


Figure 3: Dev set accuracy vs training mini batches for various compression percentages R for the SST2 dataset

	LSTM		DAN-RAND	
R(%)	Proposed	Baseline 2	Proposed	Baseline 2
10	10.71	-	1.38	-
50	9.28	9.23	0.89	0.93
70	9.21	-	0.62	-
75	9.11	9.08	0.59	0.62
90	8.46	-	0.45	-

Table 4: Inference Time on the SST2 test set (seconds)

than 16-bit. For similar compression rates (16-bit equivalent to 50% compression and 8-bit is equivalent to 75% compression) our method and baseline 2 (fixed-point-quantization) show similar inference times. Therefore, our method does not introduce any significant latency during inference while regaining the accuracy.

6 Conclusions and Future Work

In this work, we have proposed a neural model compression method based on low rank matrix factorization that can reduce the model memory footprint by an arbitrary proportion, that can be decided based on accuracy-memory trade-offs. Our method consists of compressing the model and then re-training it to recover accuracy while maintaining the reduced size. We have evaluated this approach on text classification tasks and showed that we can achieve up to 90% model size compression for both LSTM and DAN models, with minimal accuracy degradation (1%-2% relative) compared to an uncompressed model. We also showed that our method empirically outperforms common model compression baselines such as fixed point quantization and offline word embedding compression for the classification problems we examined. We have also provided an analysis of our method's effect on the model inference time and showed under which conditions it can achieve faster inference compared to model quantization techniques. An additional contribution of this work is the introduction of a novel learning rate scheduling algorithm, the Cyclically Annealed Learning Rate (CALR). We compared CALR to other popular adaptive learning rate algorithms and showed that it leads to better performance on the SST2 benchmark dataset. In future, we plan to evaluate

the proposed compression method and the proposed CALR schedule on a larger variety of NLP tasks including sequence tagging and sequence to sequence modeling.

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